

Q. Convert the Grammar into GNF (Greibach Normal Form) (7 Marks)

$$S \rightarrow ABb|a$$

$$A \rightarrow aaA|B$$

$$B \rightarrow bAb$$

⇒ Step 1: Eliminate Null Productions

→ in this example, there are no null productions.

Step 2: Elimination of Unit Productions (ie. single capital letter)

→ in this example, $A \rightarrow B$ is a unit production.

∴ Replace A with production of B

Since $B \rightarrow bAb$, substitute this into A

$$\therefore A \rightarrow aaA|bAb$$

Now, remove $A \rightarrow B$

Grammar after eliminating unit productions:

$$S \rightarrow ABb|a$$

$$A \rightarrow aaA|bAb$$

$$B \rightarrow bAb$$

Step 3: obtain GNF

→ identify GNF productions

- write GNF productions as it is which are already in GNF.

Condition for GNF: Every production must begin with a terminal: $A \rightarrow a\alpha$

where: a is a terminal

α is a sequence of non terminals

→ in this example, only one production is in GNF form.

$$\text{ie. } S \rightarrow a$$

$$S \rightarrow ABb|a$$

$$A \rightarrow aaA|bAb$$

$$B \rightarrow bAb$$

$$\rightarrow S \rightarrow \underline{aa}ABb|\underline{bA}bBb|a$$

$$A \rightarrow aaA|bAb$$

$$B \rightarrow bAb$$

} replaced productions
of A in $S \rightarrow ABb$

- since production $S \rightarrow ABb$ starts with non-terminal A, substitute productions of A, so that S begins with terminals.

$$\rightarrow S \rightarrow a\underline{C}ABD|bA\underline{D}B\underline{D}|a$$

$$C \rightarrow a$$

$$D \rightarrow b$$

$$A \rightarrow aCA|bAD$$

$$B \rightarrow bAD$$

} converted terminal
symbols into non-terminals
in production S. $\Rightarrow C \rightarrow a, D \rightarrow b$
(because every production
should starts with a
terminal symbol followed
by non-terminals only)

- In this, we introduced new variables such as C & D for terminals appearing after the first position because in GNF, only first symbol can be a terminal

* Now every production begins with a terminal followed by only non-terminals. Hence below given grammar is in GNF.

$$\rightarrow S \rightarrow aCABD|bADBDD|a$$

$$A \rightarrow aCA|bAD$$

$$B \rightarrow bAD$$

$$C \rightarrow a$$

$$D \rightarrow b$$

} GNF form (Answer)

Q. Apply the rules and show step by step conversion of the following grammar to CNF. (7 Marks)

$$S \rightarrow ASA | aB$$

$$A \rightarrow B | S$$

$$B \rightarrow b | \epsilon$$

⇒ In CNF, on the right side of CFG, only two types of length is allowed.

(i) length 1 (no of symbol is 1) : terminal symbol

(ii) length 2 (no of symbol is 2) : both should be non terminal

i.e. A production of form : $A \rightarrow BC$ or $A \rightarrow a$ } condition for CNF

where $A, B, C \in V$ and $a \in T$, is called as CNF.

Here, V = non terminal symbols (capital letters)

T = terminal symbol (small letters, special symbols)

step 1 : Eliminate Null Productions

put ϵ in place of B in every production.

$$S \rightarrow ASA | aB | a$$

$$A \rightarrow B | S | \epsilon$$

$$B \rightarrow b$$

⇒ Now put $A = \epsilon$ in the productions containing A

$$S \rightarrow ASA | SA | AS | S | aB | a$$

$$A \rightarrow B | S$$

$$B \rightarrow b$$

step 2 : Elimination of Unit Productions (ie. single capital letter)

$$A \rightarrow B$$

$$A \rightarrow S$$

$$S \rightarrow S$$

→ Remove $S \rightarrow S$

→ Replace $A \rightarrow B$

since $B \rightarrow b$,

add $A \rightarrow b$ ✓

→ Replace $A \rightarrow S$

copy productions of S : $\therefore A \rightarrow ASA | SA | AS | aB | a$ ✓

→ Grammar becomes:

$$S \rightarrow ASA \mid SA \mid AS \mid aB \mid a$$

$$A \rightarrow ASA \mid SA \mid AS \mid aB \mid a \mid b$$

$$B \rightarrow b$$

step 3: Replace terminals in mixed productions

mixed productions means terminal + non-terminal

→ in this example, mixed production is aB

∴ Introduce new Non terminal for the terminal 'a'

$$C \rightarrow a$$

Replace: $aB \rightarrow CB$

∴ Grammar becomes:

$$S \rightarrow ASA \mid SA \mid AS \mid CB \mid a$$

$$A \rightarrow ASA \mid SA \mid AS \mid CB \mid a \mid b$$

$$B \rightarrow b$$

$$C \rightarrow a$$

step 4: Convert long productions into 2 variables/non-terminals

→ in this example, long production is $S \rightarrow ASA$

we need to convert it into 2 non-terminals.

- for this, keep first non-terminal as it is, and for other two non-terminals, introduce new non-terminal.

$$S \rightarrow AD$$

here, $D \rightarrow SA$ [D is the new non-terminal]

$$S \rightarrow AD \mid SA \mid AS \mid CB \mid a$$

$$A \rightarrow AD \mid SA \mid AS \mid CB \mid a \mid b$$

$$D \rightarrow SA$$

$$B \rightarrow b$$

$$C \rightarrow a$$

as every production is of the form:

$$A \rightarrow BC \text{ or } A \rightarrow a,$$

this grammar is in CNF